Deep RL for market making

Specification

August 22, 2026

1 The question

A market maker quotes a bid and an ask around a mid price, earns the spread when someone trades against a quote, and carries the resulting inventory at risk. When the world is simple enough, the optimal quotes are known in closed form. When it is not, they are not.

This project builds a family of simulated worlds $\{\mathcal{M}_\theta\}$ with one distinguished point θ_{null} where the closed form is exactly optimal, and asks a single question:

> As the world moves away from θ_{null} along one coefficient at a time, how much does a reinforcement learning agent gain over the closed form, and which piece of information buys that gain?

The point of the design is that the answer is falsifiable. At θ_{null} the agent must *recover* the closed form and beat nothing; anywhere else, a gain is only credited when a blind agent of the same size and budget does not obtain it.

2 The criterion

Let S_t be the mid price, q_t the inventory in units, and X_t the cash. Write $W_t = X_t + q_t S_t$ for the mark-to-market wealth. Over one step of length Δt the reward is

$$r_t \;=\; \Delta W_t \;-\; \phi\, q_t^2\, \Delta t, \qquad \Delta W_t \;=\; \underbrace{\textstyle\sum \delta^{\pm}}_{\text{spread}} + \underbrace{q_t\, \Delta S_t}_{\text{carry}} + \underbrace{\Delta q_t\, \Delta S_t}_{\text{adverse}}, \tag{1}$$

where δ^+ and δ^- are the half-spreads quoted on the ask and bid side, and $\phi \geq 0$ prices inventory risk. The three-term decomposition is an identity, not an approximation, and the simulator asserts it at every step to 10^{-9}: it is what makes "the agent earned more spread" and "the agent suffered less adverse selection" separately measurable rather than rhetorical.

An episode runs for T with $n = T/\Delta t$ steps and inventory clipped to $|q| \leq Q$. The agent's objective is $\mathbb{E}\big[\sum_t r_t\big]$, undiscounted.

3 The null world and its closed form

At θ_{null} market orders arrive on each side as a Poisson process of intensity λ, a quote at distance δ is filled with probability $e^{-\kappa\delta}$, and the mid is a driftless geometric Brownian motion. Cartea and Jaimungal solve this exactly. Writing $h_q(t)$ for the value function's inventory term, the optimal half-spreads are

$$\delta^{*\pm} \;=\; \frac{1}{\kappa} \;+\; h_q \;-\; h_{q\mp 1}, \qquad \omega = e^{\kappa h} \text{ solves } \omega' = A\omega, \quad A_{q,q} = \kappa\phi q^2, \quad A_{q,q\pm 1} = -\lambda/e. \tag{2}$$

Two properties of this solution drive the whole design. It is **one-sided symmetric**: the skew depends on q and on time remaining, never on anything the market is doing. And σ does not appear in it at all — the carry term $q\,\mathrm{d}S$ is a martingale, so volatility never enters the Hamilton–Jacobi equation and inventory risk is carried entirely by ϕ.

Geometric mid. The log mid evolves as $\mathrm{d}\log S = (\alpha - \frac{1}{2}\sigma^2)\mathrm{d}t + \sigma\,\mathrm{d}W$. The Itô term is what keeps $\mathbb{E}[\mathrm{d}S] = 0$ when $\alpha = 0$; without it the mid drifts up and a long maker earns a spurious carry. Because σ multiplies the *log* increment it is a return per unit $\sqrt{\text{time}}$, dimensionless, while κ is in 1/price and δ is a price. The price level S_0 therefore carries the scale, and the exact invariance is

$$(S_0,\ \kappa,\ \phi) \longmapsto (aS_0,\ \kappa/a,\ a\phi), \qquad \sigma,\ \varepsilon,\ \zeta_{\mathrm{mkt}} \text{ unchanged}, \tag{3}$$

under which the criterion scales by a and every ratio is unchanged. This is verified numerically over a 625× range in S_0.

4 Variables

symbol	unit	meaning
State and control		
S_t	price	mid price; geometric, so strictly positive
q_t	units	inventory, clipped to $\lvert q\rvert \le Q$
$\delta^{\pm}$	price	half-spread quoted on the ask / bid side
ϕ	price/unit^2/time	inventory penalty in the criterion
Mid price and volatility		
σ	return	volatility of the log mid; 10^{-4} is one basis point
σ_σ	—	vol-of-vol of $\log\sigma$; 0 leaves σ constant
ξ_σ	1/time	mean-reversion speed of $\log\sigma$
Short-term drift (adverse selection)		
α_t	return/time	transient drift of the log mid
ζ	1/time	mean-reversion speed of α
ε	return	jump in α per informed market order
ρ	prob.	share of market orders that are informed
σ_α	return	diffusive noise of α
ζ_{mkt}	return	*permanent* log-mid jump per informed order
Order flow		
$\lambda_t^{\pm}$	1/time	market-order intensity on each side
$\bar\lambda$	1/time	baseline intensity
β	1/time	decay of the excitation
$\eta,\ \nu$	1/time	self- and cross-excitation
br	—	branching ratio $\rho(\eta+\nu)/\beta$; must stay below 1
Book depth		
$\kappa_t^{\pm}$	1/price	fill-law decay; $1/\kappa$ is the natural half-spread
ϑ	1/price	long-run level of κ
ξ	1/time	mean-reversion speed of κ
$\eta_\kappa,\ \nu_\kappa$	1/price	self- and cross-excitation of depth
Own impact		
γ_{self}	1/price	temporary depth degradation after our own fill
ζ_{perm}	return	permanent log-mid move caused by our own fill

Two diagnostics gate every world. *Economic coherence* $\sigma S_0 \mathbb{E}[\kappa]\sqrt{e/\mathbb{E}[\lambda]}$ compares the half-spread earned per round trip against the price move suffered while holding; it is expected to be of order 1, and far above it abstaining wins. *Step resolution* $\sigma S_0\sqrt{\Delta t}\,\vartheta$ is the fraction of a half-spread the price covers in one step, and must stay small or the discretisation, not the economics, sets the answer.

5 Simulator layers

Each layer adds one mechanism and can be switched off, which is what makes θ_{null} reachable exactly rather than approximately.

layer	object	off at θ_{null} when	mechanism
1	mid price	—	geometric Brownian motion
2	short-term drift	$\varepsilon = \rho = \zeta_{\text{mkt}} = 0$	flow-driven α, transient and permanent
3	order flow	$\eta = \nu = 0$	mutually exciting Hawkes intensities
4	book depth	$\eta_\kappa = \nu_\kappa = 0$	flow-driven $\kappa^\pm$
5	volatility	$\sigma_\sigma = 0$	log-OU stochastic volatility
6	own impact	$\gamma_{\text{self}} = \zeta_{\text{perm}} = 0$	our own fills move depth and mid
7	execution	—	coupled thinning of the same market orders
8	environment	—	observation, reward, episode

Layer 7 is the one that makes adverse selection exist. A market order is drawn once and then drives *both* the fill indicator $F^\pm = \mathbb{1}\{\mathrm{Bin}(n^\pm, e^{-\kappa\delta}) \geq 1\}$ and the jump in α. Drawing them separately would leave a world where getting filled tells the maker nothing, and every result about information would be an artefact.

6 Relaxation axes

One coefficient at a time, from θ_{null} outwards. Each axis names the observation block that could in principle carry its information.

axis	coefficients	mechanism	block
1	σ_σ	stochastic volatility	regime
2	η, ν	flow clustering	regime
3	ε, ρ	adverse selection, transient	signal
3d	ζ_{mkt}, ρ	adverse selection, permanent	signal
4	η_κ, ν_κ	reactive depth	book
5	k_{lat}	quote latency (a `Config` choice)	—
6	γ_{self}	own-order impact	book

Axes 1 and 2 must show the closed form *holding.* Its solution does not use σ, and it uses λ only through its level, so stochastic volatility and mean-preserving clustering should cost it little. If it collapses there, that is an implementation defect and not a discovery — which is why those two axes are run first and also serve as negative controls for the bias correction.

7 What makes a measured gain credible

The protocol is identical on every axis, and each item exists to kill one specific alternative explanation.

1. **The world is checked before the agents run.** Measured moments — $\mathbb{E}[\lambda]$, $\mathbb{E}[\kappa]$, the intensity autocorrelation, the markout curve — against their analytical targets. An axis whose moments are wrong tests nothing.
2. **Common random numbers.** Every policy in a (ϕ, seed) cell sees the same drawn world and the same market orders, so all comparisons are paired. The script verifies this itself: baseline criteria must be identical to the bit across observation layouts, or the observation is perturbing the world.

3. **Three closed forms, not one.** At the true $\bar{\lambda}$, at the stationary $\mathbb{E}[\lambda]$, and refitted on a rolling window. The first gap isolates a *level* error, which a practitioner would simply re-estimate away; what remains is blindness to the *state*, and only that part is available to an agent.
4. **A blind agent of the same size and budget.** "The agent beats the closed form" proves nothing if an agent seeing only (q, t) beats it too, since it can gain by adapting to the inventory bound rather than to the regime. The reported quantity is the informed-minus-blind gap.
5. **Two budgets.** Adding an observation block raises the parameter count and degrades the optimisation, so at equal budget an informed agent can lose for a reason unrelated to information. Each axis is run at two budgets and the gradient quality is reported alongside.
6. **Baselines other than the closed form.** A random quoter and fixed half-spreads at $0.5/\kappa$, $1/\kappa$ and $2/\kappa$, so the reader can see what the problem is worth before any optimisation.

Results are reported in financial units — basis points of notional, basis points per fill, Sharpe ratio per episode and annualised, maximum drawdown, hit rate — alongside the raw criterion, so that the size of a gain is legible without knowing the model's internal scale.

8 Deliverables

1. A layered simulator with θ_{null} reachable exactly, every layer separately switchable, and the reward identity asserted at each step.
2. An independent implementation of the Cartea–Jaimungal solution, validated at θ_{null} against a policy search over the same family.
3. Two learners — PPO and an evolution strategy — sharing the environment, the action space and the evaluation.
4. Per-axis measurement tables, with the negative controls and the bias correction they imply.
5. A calibration against public Binance data, stating which coefficients it identifies and which it cannot.
6. Figures, a notebook and an interactive dashboard reproducing the reported numbers.